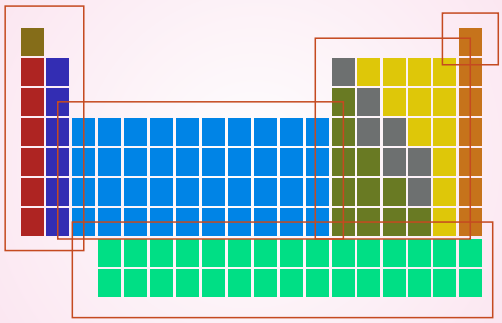
ALLEN CAREER INSTITUTE
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INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Periodic Table & Periodic Properties

(English Medium)

EXERCISE # O-1

Periodic Table

1. Which is not anomalous pair of elements in the Mendeleev's periodic table:-

(A) Ar and K (B) Co and Ni (C) Te and I (D) Al and Si

PT0001

2. Representative elements belong to :

(A) s-and p-block (B) d-block (C) d- and f-block (D) f-block

PT0002

3. True statement is :-

(A) All the transuranic elements are synthetic elements

(B) Elements of third period are called transition elements

(C) Element of [Ar] $3d^{10}4s^2$ configuration is placed in IIA group

(D) Electronic configuration of elements of a group is same

PT0003

4. Which of the following match is correct :-

(A) Last natural element – Uub

(B) General electronic configuration of IA group –ns^2

(C) All inert gas elements lies between 2nd – 6th period

(D) Typical elements – 3rd period elements

PT0004

5. The electronic configuration of elements X and Z are $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ and $1s^2\ 2s^2\ 2p^5$ respectively. What is the position of element X with respect to position of Z in the periodic table -

(A) Just below element Z (B) Just above Z

(C) Left to the Z (D) right to the Z

PT0005

6. Which of the following is not a Dobereiner triad :

(A) H, F, Cl (B) N, O, F (C) P, As, Sb (D) S, Se, Te

PT0006

7. Select the incorrect statement for Lother Meyer's curve :

(A) Curve is plotted between atomic weight and atomic volume

(B) Alkali metals occupy maxima of curve

(C) Halogens occupy descending portions of the curve

(D) Transition metals occupy bottom portions of the curve

PT0007

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Atomic & Ionic Radii

8. The size of the following species increases in the order:

(A) $Mg^{2+} < Na^{+} < F^{-}$ (B) $F^{-} < Na^{+} < Mg^{2+}$

(C) $Mg^{2+} < F^{-} < Na^{+}$ (D) $Na^{+} < F^{-} < Mg^{2+}$

PT0008

9. Highest size will be of

(A) Br^{-} (B) I (C) I^{-} (D) I^{+}

PT0009

10. Element Cu has two oxidation states Cu^{+1} & Cu^{+2}. the right order of radii of these ions.

(A) $Cu^{+1} > Cu^{+2}$ (B) $Cu^{+2} > Cu^{+1}$ (C) $Cu^{+1} = Cu^{+2}$ (D) $Cu^{+2} \geq Cu^{+1}$

PT0010

11. The correct order of increasing atomic size of element N,F, Si & P.

(A) N < F < Si < P (B) F > N < P < Si (C) F < N < P < Si (D) F < N < Si < P

PT0011

12. The correct order of atomic or ionic size

(A) N < Li < B (B) Cl < Mg < Ca (C) $Ca^{+2} < S^{-2} < Cl^{-}$ (D) $Na^{+} < Mg^{+2} < Cl^{-}$

PT0012

13. In isoelectronic series largest difference between size is observed in N^{3-}, O^{2-}, F^{-}, Na^{+}, Mg^{2+} :

(A) N^{3-}, Mg^{2+} (B) N^{3-}, O^{2-} (C) Mg^{2+}, Na^{+} (D) F^{-}, Na^{+}

PT0013

14. Mg, Mg^{2+}, Al and Al^{3+} are arranged in decreasing order of size 1 > 2 > 3 > 4. Species which are present at 1 and 4 position respectively are :

(A) Al, Mg^{2+} (B) Mg, Al^{3+} (C) Mg^{2+}, Al (D) Al^{3+}, Mg

PT0014

Ionization Energy or Potential

15. In which of the following electronic configuration, ionisation energy will be maximum in

(A) [Ne] $3s^2\ 3p^1$ (B) [Ne] $3s^2\ 3p^2$ (C) [Ne] $3s^2\ 3p^3$ (D) [Ar] $3d^{10}\ 4s^2\ 4p^3$

PT0015

16. The correct order of second ionisation potential of C, N, O and F is:

(A) C > N > O > F (B) O > N > F >C (C) O > F > N > C (D) F > O > N > C

PT0016

17. The ionization energy will be maximum for which process?

(A) Ba → Ba^{+} (B) Be → Be^{+} (C) Cs → Cs^{+} (D) Li → Li^{+}

PT0017

18. Amongst the following, the incorrect statement is

(A) IE_1 (Al) < IE_1 (Mg) (B) IE_1 (Na) < IE_1(Mg)

(C) IE_2 (Mg) > IE_2 (Na) (D) IE_3 (Mg) > IE_3 (Al)

PT0018

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19. Decreasing ionization potential for K, Ca & Ba is

(A) Ba> K > Ca (B) Ca > Ba > K

(C) K > Ba > Ca (D) K > Ca > Ba

PT0019

20. Alkaline earth metals always form dipositive ions due to

(A) $IE_2 - IE_1 > 10$ eV (B) $IE_2 - IE_1 = 17$ eV

(C) $IE_2 - IE_1 < 10$ eV (D) None of these

PT0020

21. The correct order of second I.P.

(A) Na < Mg > Al < Si (B) Na > Mg < Al > Si

(C) Na > Mg > Al < Si (D) Na > Mg > Al > Si

PT0021

Electron affinity or Electron Gain Enthalpy

22. The process requires absorption of energy is

(A) $F \rightarrow F^-$ (B) $Cl \rightarrow Cl^-$ (C) $O^- \rightarrow O^{2-}$ (D) $H \rightarrow H^-$

PT0022

23. Of the following elements, which possesses the highest electron affinity?

(A) As (B) O (C) S (D) Se

PT0023

24. Electron affinities of O,F,S and Cl are in the order.

(A) O < S < Cl < F (B) O < S < F < Cl

(C) S < O < Cl < F (D) S < O < F < Cl

PT0024

25. Increasing order of Electron affinity for following configuration.

(a) $1s^2, 2s^2\ 2p^2$ (b) $1s^2, 2s^2\ 2p^4$

(c) $1s^2, 2s^2\ 2p^6\ 3s^2\ 3p^4$ (d) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^3$

(A) d < a < b < c (B) d < a < c < b (C) a < b < c < d (D) a < b < d < c

PT0025

26. Highest electron affinity is shown by

(A) F^- (B) Cl^- (C) Li^+ (D) Na^+

PT0026

27. Which of the following statements is not true?

(A) F atom can hold additional electron more tightly than Cl atom

(B) Cl atom can hold additional electron more tightly than F atom

(C) The incoming electron encounters greater repulsion for F atom than for Cl atom

(D) It is easier to remove an electron from F^- than Cl^-.

PT0027

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Electronegativity

28. The outermost electronic configuration of most electronegative element amongst the following is :
(A) $ns^2\ np^3$ (B) $ns^2\ np^4$ (C) $ns^2\ np^5$ (D) $ns^2\ np^6$

PT0028

29. In the following, which is the correct representation ?
(A) $\overset{\delta+}{C}-\overset{\delta-}{F}$ (B) $\overset{\delta+}{C}-\overset{\delta+}{Cl}$ (C) $\overset{\delta+}{F}-\overset{\delta-}{Cl}$ (D) $\overset{\delta-}{O}-\overset{\delta+}{F}$

PT0029

30. On the Pauling's electronegativity scale, which element is next to F .
(A) Cl (B) O (C) Br (D) Ne

PT0030

31. Which one is not correct order of electronegativity.
(A) F > Cl > Br > I (B) Si > Al > Mg > Na
(C) Cl > S > P > Si (D) None of these

PT0031

32. The increasing order of acidic nature of Li_2O, BeO, B_2O_3
(A) $Li_2O > BeO < B_2O_3$ (B) $Li_2O < BeO < B_2O_3$
(C) $Li_2O < BeO > B_2O_3$ (D) $Li_2O > BeO > B_2O_3$

PT0032

33. The lowest electronegativity of the element from the following atomic number is.
(A) 37 (B) 55 (C) 9 (D) 35

PT0033

Miscellaneous

34. Which of the following does not reflect the periodicity of element
(A) Bonding behaviour (B) Electronegativity
(C) Ionisation potential (D) Neutron/ Proton ratio

PT0034

35. Among the following, which species is/are paramagnetic ?
(i) Sr^{2+} (ii) Fe^{3+} (iii) Co^{2+} (iv) S^{2-} (v) Pb^{2+}
(A) i, iv, v (B) i, ii, iii (C) ii, iii (D) iv, v

PT0035

36. Choose the s-block element from the following:
(A) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5, 4s^1$ (B) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^1$
(C) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$ (D) all of the above

PT0036

37. False statement for periodic classification of elements is
(A) The properties of the elements are periodic function of their atomic numbers.
(B) No. of non-metallic elements is less than the no. of metallic elements.
(C) First ionization energy of elements does not increase regularly with the increase in atomic number in a period.
(D) d-subshell is filled by final electron with increasing atomic number of inner transition elements.

PT0037

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38. Which of the following order is incorrect against the property indicated :
(A) Mg < Ar < Na (2nd I.E.) (B) Be < F < Cl ($|\Delta H_{eg}|$)
(C) Rb < Na < K > Ca (atomic radius) (D) P < S < N (electronegativity)

PT0038

39. If each orbital can hold a maximum of three electrons, the number of elements in 9th period of periodic table (long form) will be
(A) 48 (B) 162 (C) 50 (D) 75

PT0039

40. Which of the following element has highest metallic character .

	Element	**IP**
(A)	P	17 eV
(B)	Q	2 eV
(C)	R	10 eV
(D)	S	13 eV

PT0040

41. The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$. The atomic number and the group number of the element 'X' which is just below the above element in the periodic table are respectively.
(A) 24 & 6 (B) 24 & 15 (C) 34 & 16 (D) 34 & 8

PT0041

42. The number of d- electrons in Mn^{2+} is equal to that of
(A) p-electrons in N (B) s-electrons in Na
(C) d-electrons in Fe^{+2} (D) p-electrons in O^{-2}

PT0042

43. Which of the following has the maximum number of unpaired electrons
(A) Mg^{2+} (B) Ti^{3+} (C) V^{3+} (D) Fe^{2+}

PT0043

44. EN of the element (A) is E_1 and IP is E_2. Hence EA will be according to mulliken
(A) $2E_1 - E_2$ (B) $E_1 - E_2$ (C) $E_1 - 2E_2$ (D) $(E_1 + E_2)/2$

PT0044

45. Moving from right to left in a periodic table, the atomic size is:
(A) Increased (B) Decreased (C) Remains constant (D) None of these

PT0045

46. One element has atomic weight 39. Its electronic configuration is $1s^2$, $2s^2\ 2p^6$, $3s^2\ 3p^6\ 4s^1$. The true statement for that element is:
(A) High value of IE (B) Transition element (C) Isotone with $_{18}Ar^{38}$ (D) None

PT0046

47. The number of paired electrons in oxygen atom is:
(A) 6 (B) 16 (C) 8 (D) 32

PT0047

48. The decreasing size of K^+, Ca^{2+}, Cl^- & S^{2-} follows the order:
(A) $K^+ > Ca^{+2} > S^{-2} > Cl^-$ (B) $K^+ > Ca^{+2} > Cl^- > S^{-2}$
(C) $Ca^{+2} > K^+ > Cl^- > S^{-2}$ (D) $S^{-2} > Cl^- > K^+ > Ca^{+2}$

PT0048

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EXERCISE # O-2

Atomic & Ionic Radius

1. Select correct order of size :

(A) $Ti^{2+} < Ti < Zr$ (B) $Ti^{2+} < Ti < Hf$ (C) $Zr^{2+} < Zr \approx Hf$ (D) $Hf^{2+} < Hf \approx Zr$

PT0049

2. Which of the following orders of atomic / Ionic radius is correct ?

(A) B < Al > Ga (B) Sc > Cu < Zn (C) C < O < N (D) $Al^{+3} < Al^{+2} < Al^{+}$

PT0050

Electron Affinity

3. Which of the following is correct order of EA.

(A) N < C < O < F (B) F > Cl > Br > I (C) Cl > F > Br > I (D) C < N < O < F

PT0051

4. The electron affinity of the members of oxygen family of the periodic table, follows the sequence

(A) O > S > Se (B) S > O < Se (C) O < S > Se (D) Se > O > S

PT0052

Ionisation Energy

5. Considering the following ionisation steps :

$A(g) \rightarrow A^{+}(g) + e^{-}$ $\Delta H = 100$ eV $A(g) \rightarrow A^{2+}(g) + 2e^{-}$ $\Delta H = 250$ eV

Select the correct statements :

(A) IE_1 of A(g) is 100 eV (B) IE_1 of $A^{+}(g)$ is 150 eV

(C) IE_2 of A(g) is 150 eV (D) IE_2 of A(g) is 250 eV

PT0053

6. Which of the following are correct ?

(A) $IE_2(Mg) < IE_2(Na)$ (B) EA (N) < EA (P)

(C) Atomic size Mg^{+2} > Atomic size (Li^{+}) (D) IP of Na < Mg < Al

PT0054

Electronegativty

7. Amongst the following statements, which is / are correct?

(A) Electronegativity of sulphur is greater than that of oxygen.

(B) Electron affinity of oxygen is smaller than that of sulphur.

(C) Electron gain enthalpy of fluorine is most negative

(D) Electron gain enthalpy of chlorine is most negative

PT0055

8. An element 'E' have IE = x eV/atom and EA = y eV / atom and EN on Pauling scale is 1.2. Find EN of 'E' on Mullikan scale :

(A) $\frac{x+y}{2}$ (B) 1.2×2.8 (C) $\frac{x+y}{2.8} - 1.2$ (D) $1.2 - \frac{x+y}{2}$

PT0056

Hydration Energy

9. Choose the INCORRECT order of hydrated size of the ions -

(A) $F^{\ominus}_{(aq.)} > Cl^{\ominus}_{(aq.)} > Br^{\ominus}_{(aq.)} > I^{\ominus}_{(aq.)}$ (B) $Rb^{\oplus}_{(aq.)} > K^{\oplus}_{(aq.)} > Na^{\oplus}_{(aq.)} > Li^{\oplus}_{(aq.)}$

(C) $Na^{\oplus}_{(aq.)} > Mg^{2+}_{(aq.)} > Al^{3+}_{(aq.)}$ (D) $Be^{2+}_{(aq.)} > Mg^{2+}_{(aq.)} > Ca^{2+}_{(aq.)} > Sr^{2+}_{(aq.)}$

PT0057

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10. Find the correct ionic mobility order in aqueous solution from the following options-

(A) $Li^+ < Na^+$ (B) $Mg^{2+} < Sr^{2+}$ (C) $Na^+ < K^+$ (D) $F^- < Cl^-$

PT0058

Miscellaneous

11. Select the correct statement(s).

(A) The value of electron gain enthalpy of an element can be -ve or +ve.

(B) In the periodic table, metallic character of the elements increases down the group and decreases across the period

(C) The Cl^- & S^{2-} are isoelectronic species but first one is not smaller in size than the second

(D) Ionization enthalpy of an atom is equal to electron gain enthalpy of cation

PT0059

12. In halogens, which of the following properties increase from iodine to fluroine

(A) Ionisation energy (B) Electronegativity

(C) Bond length (D) Electron affinity

PT0060

13. In which of the following set of elements 1st element is more metallic than second.

(A) Ba, Ca (B) Sb, Sn (C) Ge, S (D) Na, F

PT0061

14. Which of the following order(s) is / are **CORRECT** :

(A) Li < Be < B < C (IE_1)

(B) HF < HCl < HBr < HI (Bond length)

(C) $Na_2O < MgO < Al_2O_3 < SiO_2 < P_2O_5$ (Acidic)

(D) $Li^+(g) < Na^+(g) < K^+(g) < Cs^+(g)$ (Ionic radius)

PT0062

15. Which of the following order is correct :

(A) P < Si < Be < Mg < Na (Metallic character)

(B) $Mg^{+2} < Na^+ < F^- < O^{2-}$ (Ionic radius)

(C) Li < B < Be < C < N < O (2nd ionization energy)

(D) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$ (Ionic mobility in infinite dilute aqueous solution)

PT0063

16. The ionic compound $A^+ B^-$ is formed easily when the

(A) electron affinity of B is high (B) ionization energy of A is low

(C) lattice energy of AB is high (D) lattice energy of AB is low

PT0064

17. Which of the following is/are correct?

(A) For $A(g) + e^- \longrightarrow A^-(g)$ ΔH may be negative

(B) For $A^-(g) + e^- \longrightarrow A^{2-}(g)$ ΔH may be negative

(C) For $A^-(g) + e^- \longrightarrow A^{2-}(g)$ ΔH must be positive

(D) For $Ne(g) + e^- \longrightarrow Ne^-(g)$ ΔH may be zero

PT0065

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EXERCISE # S-1

1. Find out the atomic number of element whose IUPAC name is Unnilpentium.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

PT0066

2. Select total number of acidic compounds out of given below.

CsOH, $OC(OH)_2$, $SO_2(OH)_2$, $Sr(OH)_2$, $Ca(OH)_2$, $Ba(OH)_2$, BrOH, NaOH, O_2NOH

PT0067

3. Total number of enthalpy(s) (out of given eight) of A(g) which is/are **not** associated with conversion of $A^{-}_{(g)} \longrightarrow A^{+4}_{(g)}$:

IE_1, IE_2, IE_3, IE_4, IE_5, IE_6, EGE_1, EGE_2

(IE = ionization energy, EGE = electron gain enthalpy)

PT0068

4. Write the number of pairs in which size of first element or ion is higher as compared to IInd out of following eight pairs.

(O,S) , (He, Ne) , (Kr, Ne), (Na, Na^+), (Cl, Cl^-), (I^- , Cl^-), (Li, Na), (Li^+ , Na^+)

PT0069

5. Total number of elements which have less IE_1 than that of 'N'.

Be, B, C, F, P, He

PT0070

6. Size of H^- is smaller than how many elements among these ?

H, Li, Be, B, C, N, O, F, F^-

PT0071

7. How many orders are **CORRECT**

(a) Acidic strength $\Rightarrow$ ClOH < BrOH < IOH

(b) Basic strength $\Rightarrow$ MgO < CaO < SrO < BeO

(c) Electronegativity $\Rightarrow$ I < Br < N < O < F

(d) Electron affinity $\Rightarrow$ Mg < Na < Si < S < Cl

(e) % Ionic character $\Rightarrow$ NaF < KF < RbF

PT0072

8. _____ is higher for fluorine as compared to chlorine.

Find the number of properties given below to fill the blank space to make a correct statement.

Atomic mass, Covalent radius, Ionic radius (X^-), Ionization energy,Electron affinity, Electronegativity, Hydration energy of uninegative ion (X^-),

PT0073

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9. Upto argon find the number of elements which have lower IE_1 as compared to He.

PT0074

10. The number of pairs, in which EA of the second element is more than that of the first element is :
[O, S], [C, N], [O, N], [N, P], [Cl^+, F^+], [K^+, Na^+]

PT0075

11. For an element the successive ionisation energy values (in eV/atom), are given below.
14.534, 29.601, 47.448, 77.472, 97.888, 552.057, 667.029
Find the number of valence shell electrons in that element.

PT0076

Subjective :

12. Calculate E.N. of chlorine atom on Pauling scale if I.E. of Cl^- is 4eV & of E.A. of Cl^+ is + 13.0 eV.

PT0077

13. Increasing order of ionic size :
N^{3-}, Na^+, F^-, O^{2-}, Mg^{2+}

PT0078

14. How many of the following have higher lattice energy than KI ?
LiI, LiCl, KCl, NaI, LiBr, MgI_2, NaCl, AlF_3, K_2O, CsI, $CaBr_2$, RbI.

15. How many of the following ions have higher hydration energy than $Ca^{2+}_{(aq)}$?
Ba^{2+}, Be^{2+}, Cs^+, K^+, Mg^{2+}, Rb^+

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EXERCISE # S-2

Paragraph for Questions 1 to 2

First electron gain enthalpy (in $\frac{kJ}{mol}$) of few elements are given below :

Elements	ΔH_{eg}
I	–60
II	–45
III	–328
IV	–295
V	+ 48

Answer the following questions on the basis of above data:

1. Which element may be an inert gas

(A) I (B) III (C) IV (D) V

PT0079

2. Which element is most non-metallic among all the elements -

(A) I (B) II (C) III (D) IV

PT0079

Paragraph for Questions 3 and 4

The IE_1 and the IE_2 in KJ/mol of a few elements designated by U, V, W, X are shown below.

Atom	IE_1	IE_2
U	2464	6110
V	610	7542
W	928	1810
X	1588	3410

Based on the above information answer the follwoing question :-

3. Which of the elements represent a noble gas.

(A) U (B) V (C) W (D) X

PT0080

4. Which of the following element belongs to group 1 (IA).

(A) U (B) V (C) W (D) X

PT0080

Paragraph for Question 5 to 7

Nature of bond can be predicted on the basis of electronegativity of bonded atoms, greater difference in electronegativity (X), more will be the polarity of bond, and polar bond are easily broken in polar solvent like water. For hydroxy acids $X_O - X_A$ difference predict the nature of oxide formed by the element A.

$|X_O - X_A| > |X_O - X_H|$ then A–O–H show basic nature (NaOH)

$|X_O - X_A| < |X_O - X_H|$ then A–O–H show acidic nature (H–O–Cl)

With the help of EN values [$EN_A = 1.8$, $EN_B = 2.6$, $EN_C = 1.6$, $EN_D = 2.8$] answer the following questions for the compounds HAO, HBO, HCO, HDO.

5. Compounds whose aqueous solution is acidic and order of their acidic strength

(A) AOH, COH ; AOH < COH (B) HDO, HBO ; HDO > HBO

(C) AOH, COH ; AOH > COH (D) HDO, HBO ; HDO < HBO

PT0081

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6. Compounds whose aqueous solution is basic and order of their basic strength

(A) AOH, COH ; AOH < COH (B) HDO, HBO ; HDO > HBO

(C) AOH, COH ; AOH > COH (D) HDO, HBO ; HDO < HBO

PT0081

7. Percentage ionic character of compound AB is

(A) 42.42 % (B) 24.24 % (C) 15.04 % (D) None of these

PT0081

Paragraph for Question 8 to 12

All the simple salt dissolve in water, producing ions and consequently the solution conduct electricity. In this process water molecule surround both the cations and anions & release energy. This process is called hydration & energy released is called hydration energy & it depends on size of gaseous ions.

Answer the following questions with respect to given cations.

Na^+, Mg^{2+}, Al^{3+}

8. Order of extent of hydration

(A) $Na^+ = Mg^{2+} = Al^{3+}$ (B) $Na^+ > Mg^{2+} > Al^{3+}$

(C) $Al^{+3} > Mg^{2+} > Na^+$ (D) $Al^{+3} > Mg^{2+} < Na^+$

PT0082

9. Order of hydration energy

(A) $Na^+ = Mg^{2+} = Al^{3+}$ (B) $Na^+ > Mg^{2+} > Al^{3+}$

(C) $Al^{+3} > Mg^{2+} > Na^+$ (D) $Al^{+3} > Mg^{2+} < Na^+$

PT0082

10. Order of size of hydrated ion.

(A) $Na^+ = Mg^{2+} = Al^{3+}$ (B) $Na^+ > Mg^{2+} > Al^{3+}$

(C) $Al^{+3} > Mg^{2+} > Na^+$ (D) $Al^{+3} > Mg^{2+} < Na^+$

PT0082

11. Order of ionic mobility

(A) $Na^+ = Mg^{2+} = Al^{3+}$ (B) $Na^+ > Mg^{2+} > Al^{3+}$

(C) $Al^{+3} > Mg^{2+} > Na^+$ (D) $Al^{+3} > Mg^{2+} < Na^+$

PT0082

12. Order of size of gaseous ions.

(A) $Na^+ = Mg^{2+} = Al^{3+}$ (B) $Na^+ > Mg^{2+} > Al^{3+}$

(C) $Al^{+3} > Mg^{2+} > Na^+$ (D) $Al^{+3} > Mg^{2+} < Na^+$

PT0082

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Paragraph for Question 13 to 14 are based on the following information.

Four elements P, Q, R & S have ground state electronic configuration as:

P → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ Q → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

R → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^3$ S → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^1$

13. Comment which of the following option represent the correct order of true (T) & false (F) statement.

I size of P < size of Q

II size of R < size of S

III size of P < size of R (appreciable difference)

IV size of Q < size of S (appreciable difference)

(A) TTTT (B) TTTF (C) FFTT (D) TTFF

PT0083

14. Order of IE_1 values among the following is

(A) P > R > S > Q (B) P < R < S < Q (C) R > S > P > Q (D) P > S > R > Q

PT0083

Matching List :

15.

Column-I **Element**		**Column-II** **Period and group number respectively**
(P)	Si	(1) 4, 3
(Q)	Sc	(2) 3, 14
(R)	Ga	(3) 6, 13
(S)	Tℓ	(4) 4, 13

Code :

	P	**Q**	**R**	**S**
(A)	2	1	3	4
(B)	1	2	4	3
(C)	2	1	4	3
(D)	4	3	1	2

PT0084

Match the column :

16. Match the column :

Column-I	**Column-II**
(A) Highest density	(P) Lithium
(B) Metallic character	(Q) Osmium
(C) Lightest Metal	(R) Mercury
(D) Liquid at room temperature	(S) Bromine

PT0085

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17. If electrons are filled in the sub shells of an atom in the following order 1s, 2s, 2p, 3s, 3p, 3d, 4s, 4p, 4d, 4f..... then match the following element in List I with block in List II.

List-I	**List-II**
(A) K(19)	(P) s-Block
(B) Fe(26)	(Q) p-Block
(C) Ga(31)	(R) d-Block
(D) Sn(50)	(S) f-block

PT0086

18. Match the characteristics mentioned in List II with the process in List I.

List-I	**List-II**
(A) $O(g) + e^- \rightarrow O^-(g)$	(P) Positve electron gain enthalpy
(B) $O^-(g) + e^- \rightarrow O^{2-}(g)$	(Q) Negative electron gain enthalpy
(C) $Na^-(g) \rightarrow Na(g) + e^-$	(R) Exothermic
(D) $Mg^+(g) + e^- \rightarrow Mg(g)$	(S) Endothermic

PT0087

19. Match the column :

Column-I	**Column-II**
(A) Cl	(P) Metal
(B) F	(Q) Highest negative electron gain enthalpy
(C) Cu	(R) Most Electronegative element
(D) He	(S) Highest ionisation energy.

PT0088

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EXERCISE # JEE-MAIN

1. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be :- **[JEE-MAIN-2013]**

(1) – 2.55 eV (2) – 5.1 eV (3) – 10.2 eV (4) + 2.55 eV

PT0092

2. Given **[JEE-MAIN 2013 (On-Line)]**

Reaction	**Energy Change (in kJ)**
$Li(s) \longrightarrow Li(g)$	161
$Li(g) \longrightarrow Li^+(g)$	520
$\frac{1}{2}F_2(g) \longrightarrow F(g)$	77
$F(g) + e^- \longrightarrow F^-(g)$	(Electron gain enthalpy)
$Li^+(g) + F^-(g) \longrightarrow LiF(s)$	–1047
$Li(s) + \frac{1}{2}F_2(g) \longrightarrow Li\,F(s)$	–617

Based on data provided, the value of electron gain enthalpy of fluorine would be :

(1) –300 kJ mol^{-1} (2) –328 kJ mol^{-1} (3) –350 kJ mol^{-1} (4) –228 kJ mol^{-1}

PT0094

3. The order of increasing sizes of atomic radii among the elements O, S, Se and As is : **[JEE-MAIN 2013 (On-Line)]**

(1) As < S < O < Se (2) O < S < As < Se

(3) Se < S < As < O (4) O < S < Se < As

PT0095

4. Which is the correct order of second ionization potential of C, N, O and F in the following ? **[JEE-MAIN 2013 (On-Line)]**

(1) O > F > N > C (2) O > N > F > C

(3) C > N > O > F (4) F > O > N > C

PT0096

5. Which of the following series correctly represents relations between the elements from X to Y ? **[JEE-MAIN 2014 (On-Line)]**

X ⟶ Y

(1) $_{18}Ar \rightarrow {}_{54}Xe$ Noble character increases

(2) $_{3}Li \rightarrow {}_{19}K$ Ionization enthalpy increases

(3) $_{6}C \rightarrow {}_{32}Ge$ Atomic radii increases

(4) $_{9}F \rightarrow {}_{35}Br$ Electron gain enthalpy with negative sign increases

PT0097

6. The ionic radii (in Å) of N^{3-}, O^{2-} and F^- are respectively :- **[JEE-MAIN 2015 (Off-Line)]**

(1) 1.71, 1.40 and 1.36 (2) 1.71, 1.36 and 1.40

(3) 1.36, 1.40 and 1.71 (4) 1.36, 1.71 and 1.40

PT0098

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7. In the long form of the periodic table, the valence shell electronic configuration of $5s^2\ 5p^4$ corresponds to the element present in: **[JEE-MAIN 2015 (On-Line)]**

(1) Group 16 and period 5 (2) Group 17 and period 6

(3) Group 17 and period 5 (4) Group 16 and period 6

PT0099

8. Which of the following atoms has the highest first ionization energy ?

[JEE-MAIN 2016 (Off-Line)]

(1) Sc (2) Rb (3) Na (4) K

PT0100

9. The non-metal that does not exhibit positive oxidation state is :

[JEE-MAIN 2016 (On-Line)]

(1) Oxygen (2) Fluorine (3) Iodine (4) Chlorine

PT0101

10. The electronic configuration with the highest ionization enthalpy is:- **[JEE-MAIN 2017]**

(1) [Ar] $3d^{10}4s^2\ 4p^3$ (2) [Ne] $3s^2\ 3p^1$

(3) [Ne] $3s^2\ 3p^2$ (4) [Ne] $3s^2\ 3p^3$

PT0102

11. Consider the following ionization enthalpies of two elements 'A' and 'B' :

Element	Ionization enthalpy (kJ/mol)		
	1st	2nd	3rd
A	899	1757	14847
B	737	1450	7731

[JEE-MAIN 2017]

Which of the following statements is correct ?

(1) Both A and B belong to group-2 where A comes below B

(2) Both A and B belong to group-1 where A comes below B

(3) Both A and B belong to group-1 where B comes below A

(4) Both A and B belong to group-2 where B comes below A

PT0103

12. In the following reactions, ZnO is respectively acting as a/an : **[JEE-MAIN 2017]**

(a) $ZnO + Na_2O \rightarrow Na_2ZnO_2$

(b) $ZnO + CO_2 \rightarrow ZnCO_3$

(1) base and acid (2) base and base (3) acid and acid (4) acid and base

PT0104

13. The group having isoelectronic species is :- **[JEE-MAIN 2017]**

(1) O^{2-} , F^-, Na^+ , Mg^{2+} (2) O^-, F^-, Na , Mg^+

(3) O^{2-} , F^-, Na , Mg^{2+} (4) O^- , F^-, Na^+ , Mg^{2+}

PT0105

14. The correct order of electron affinity is :- **[JEE-MAIN 2018]**

(1) Cl > F > O (2) F > O > Cl (3) F > Cl > O (4) O > F > Cl

PT0106

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15. For Na^+, Mg^{2+}, F^- and O^{2-} ; the correct order of increasing ionic radii is : **[JEE-MAIN 2018]**

(1) $Mg^{2+} < Na^+ < F^- < O^{2-}$ (2) $O^{2-} < F^- < Na^+ < Mg^{2+}$

(3) $Na^+ < Mg^{2+} < F^- < O^{2-}$ (4) $Mg^{2+} < O^{2-} < Na^+ < F^-$

PT0107

16. The amphoteric hydroxide is : **[JEE-MAIN ONLINE 2019]**

(1) $Ca(OH)_2$ (2) $Be(OH)_2$ (3) $Sr(OH)_2$ (4) $Mg(OH)_2$

PT0108

17. The correct option with respect to the Pauling electronegativity values of the elements is :-

(1) Ga < Ge (2) Si < Al **[JEE-MAIN ONLINE 2019]**

(3) P > S (4) Te > Se

PT0109

18. The effect of lanthanoid contraction in the lanthanoid series of elements by and large means :

(1) decrease in both atomic and ionic radii **[JEE-MAIN ONLINE 2019]**

(2) increase in atomic radii and decrease in ionic radii

(3) increase in both atomic and ionic radii

(4) decrease in atomic radii and increase in ionic radii

PT0110

19. When the first electron gain enthalpy ($\Delta_{eg}H$) of oxygen is –141 kJ/mol, its second electron gain enthalpy is : **[JEE-MAIN ONLINE 2019]**

(1) almost the same as that of the first

(2) negative , but less negative than the first

(3) a positive value

(4) a more negative value than the first

PT0111

20. The correct order of hydration enthalpies of alkali metal ions is - **[JEE-MAIN ONLINE 2019]**

(1) $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

(2) $Li^+ > Na^+ > K^+ > Cs^+ > Rb^+$

(3) $Na^+ > Li^+ > K^+ > Rb^+ > Cs^+$

(4) $Na^+ > Li^+ > K^+ > Cs^+ > Rb^+$

PT0112

21. The IUPAC symbol for the element with atomic number 119 would be : **[JEE-MAIN ONLINE 2019]**

(1) unh (2) uun (3) une (4) uue

PT0113

22. The element having greatest difference between its first and second ionization energies, is : **[JEE-MAIN ONLINE 2019]**

(1) Ca (2) K (3) Ba (4) Sc

PT0114

23. The correct statements among I to III regarding group 13 element oxides are,

(I) Boron trioxide is acidic. **[JEE-MAIN ONLINE 2019]**

(II) Oxides of aluminium and gallium are amphoteric.

(III) Oxides of indium and thallium are basic.

(1) (I), (II) and (III) (2) (II) and (III) only

(3) (I) and (III) only (4) (I) and (II) only

PT0115

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24. Consider the hydrates ions of Ti^{2+}, V^{2+}, Ti^{3+} and Sc^{3+}. The correct order of their spin-only magnetic moments is : **[JEE-MAIN ONLINE 2019]**

(1) $Sc^{3+} < Ti^{3+} < Ti^{2+} < V^{2+}$ (2) $Ti^{3+} < Ti^{2+} < Sc^{3+} < V^{2+}$

(3) $Sc^{3+} < Ti^{3+} < V^{2+} < Ti^{2+}$ (4) $V^{2+} < Ti^{2+} < Ti^{3+} < Sc^{3+}$

PT0116

25. The pair that has similar atomic radii is : **[JEE-MAIN ONLINE 2019]**

(1) Sc and Ni (2) Ti and Hf (3) Mo and W (4) Mn and Re

PT0117

26. Within each pair of elements of F & Cl , S & Se, and Li & Na, respectively, the elements that release more energy upon an electron gain are- **[JEE-MAIN ONLINE 2020]**

(1) F, Se and Na

(2) F, S and Li

(3) Cl, S and Li

(4) Cl, Se and Na

PT0118

27. The atomic radius of Ag is closest to : **[JEE-MAIN ONLINE 2020]**

(1) Cu (2) Hg (3) Au (4) Ni

PT0119

28. The third ionization enthalpy is minimum for : **[JEE-MAIN ONLINE 2020]**

(1) Fe (2) Ni (3) Co (4) Mn

PT0120

29. B has a smaller first ionization enthalpy than Be. Consider the following statements :

(I) It is easier to remove 2p electron than 2s electron

(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be.

(III) 2s electron has more penetration power than 2p electron.

(IV) atomic radius of B is more than Be

(Atomic number B = 5, Be = 4)

The correct statements are : **[JEE-MAIN ONLINE 2020]**

(1) (I), (II) and (III) (2) (II), (III) and (IV)

(3) (I), (III) and (IV) (4) (I), (II) and (IV)

PT0121

30. Lattice enthalpy and enthalpy of solution of NaCl are 788 kJ mol^{-1} and 4 kJ mol^{-1}, respectively. The hydration enthalpy of NaCl is : **[JEE MAINS 2020]**

(1) –780 kJ mol^{-1} (2) –784 kJ mol^{-1} (3) 780 kJ mol^{-1} (4) 784 kJ mol^{-1}

PT0135

31. Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is : **[JEE MAINS 2020]**

(1) Z < Y < X (2) X < Z < Y (3) X < Y < Z (4) Y < X < Z

PT0136

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32. The ionic radius of Na^+ ions is 1.02 Å. The ionic radii (in Å) of Mg^{2+} and Al^{3+}, respectively, are- **[JEE MAINS 2021]**

(1) 1.05 and 0.99

(2) 0.72 and 0.54

(3) 0.85 and 0.99

(4) 0.68 and 0.72

PT0137

33. The correct order of most negative electron gain enthalpy is **[JEE MAINS 2021]**

(1) S > Se > Te > O (2) Te > Se > S > O (3) O > S > Se > Te (4) S > O > Se > Te

PT0138

34. The total number of acidic oxides from the following list is : **[JEE MAINS 2022]**

NO, N_2O, B_2O_3, N_2O_5, CO, SO_3, P_4O_{10}

(A) 3 (B) 4 (C) 5 (D) 6

PT0143

35. The IUPAC nomenclature of an element with electronic configuration $[Rn]5f^{14}6d^17s^2$ is : **[JEE MAINS 2022]**

(1) Unnilbium

(2) Unnilunium

(3) Unnilquadium

(4) Unniltrium

PT0144

36. Outermost electronic configurations of four elements A, B, C, D are given below: **[JEE MAINS 2022]**

(A) $3s^2$ (B) $3s^2\ 3p^1$ (C) $3s^2\ 3p^3$ (D) $3s^2\ 3p^4$

PT0145

The correct order of first ionization enthalpy for them is:

(A) (A) < (B) < (C) < (D) (B) (B) < (A) < (D) < (C)

(C) (B) < (D) < (A) < (C) (D) (B) < (A) < (C) < (D)

37. An element A of group 1 shows similarity to an element B belonging to group 2. If A has maximum hydration enthalpy in group 1 then B is : **[JEE MAINS 2022]**

(A) Mg (B) Be (C) Ca (D) Sr

PT0146

38. The incorrect statement is **[JEE MAINS 2022]**

(A) The first ionization enthalpy of K is less than that of Na and Li

(B) Xe does not have the lowest first ionization enthalpy in its group

(C) The first ionization enthalpy of element with atomic number 37 is lower than that of the element with atomic number 38.

PT0147

(D) The first ionization enthalpy of Ga is higher than that of the d-block element with atomic number 30.

39. The spin-only magnetic moment value of the most basic oxide of vanadium among V_2O_3, V_2O_4 and V_2O_5 is _____ B.M. (Nearest Integer)

PT0148

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EXERCISE # JEE-ADVANCED

1. **Statement-1 :** F atom has a less negative electron gain enthalpy than Cl atom. **[JEE 2000]**
Statement-2 : Additional electron is repelled more efficiently by 3p electron in Cl atom than by 2p electron in F atom.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

PT0122

2. The correct order of radii is: **[JEE 2000]**
(A) N < Be < B (B) $F^- < O^{2-} < N^{3-}$ (C) Na < Li < K (D) $Fe^{3+} < Fe^{2+} < Fe^{4+}$

PT0123

3. The IE_1 of Be is greater than that of B. **[T/F]** **[JEE 2001]**

PT0124

4. The set representing correct order of IP_1 is **[JEE 2001]**
(A) K > Na > Li (B) Be > Mg > Ca (C) B > C > N (D) Fe > Si > C

PT0125

5. Identify the least stable ion amongst the following: **[JEE 2002]**
(A) Li^- (B) Be^- (C) B^- (D) C^-

PT0126

6. The increasing order of atomic radii of the following group 13 elements is **[JEE 2016]**
(A) Al < Ga < In < Tl (B) Ga < Al < In < Tl
(C) Al < In < Ga < Tl (D) Al < Ga < Tl < In

PT0127

7. The option(s) with only amphoteric oxides is (are): **[JEE 2017]**
(A) Cr_2O_3, CrO, SnO, PbO (B) NO, B_2O_3, PbO, SnO_2
(C) Cr_2O_3, BeO, SnO, SnO_2 (D) ZnO, Al_2O_3, PbO, PbO_2

PT0128

8. The 1st , 2nd and the 3rd ionization enthalpies I_1, I_2 and I_3, of four atoms with atomic numbers n, n+1, n+2 and n+3 , where n < 10, are tabulated below. What is the value of n? **[JEE 2020]**

Atomic number	Ionization Enthalpy (kJ/mol)		
	I_1	I_2	I_3
n	1681	3374	6050
n + 1	2081	3952	6122
n + 2	496	4562	6910
n + 3	738	1451	7733

PT0139

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ANSWER-KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	A	D	A	B	C	A	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	B	C	C	B	C	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	C	B	A	C	A	C	A	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	B	D	C	C	D	C	D	B
Que.	41	42	43	44	45	46	47	48		
Ans.	C	B	D	A	A	C	A	D		

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,B,D	A,C	B,C	A,B,C	A,B	B,D	A,B	B,C	A,B,C,D
Que.	11	12	13	14	15	16	17			
Ans.	A,B,D	A,B	A,C,D	B,C,D	A,B,D	A,B,C	A,C			

EXERCISE # S-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	6	4	3	3	4	0	3	3	17	4
Que.	11	12	13			14	15			
Ans.	5	3.03 (Pauling)	$Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$			10.00	2.00			

EXERCISE # S-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	A	B	B	A	C	C	C	C
Que.	11	12	13	14	15	16				
Ans.	B	B	B	A	C	(A)→Q;(B)→P,Q,R ;(C)→P;(D)→R, S				
Que.	17					18				
Ans.	(A)→R ; (B)→R ; (C)→Q ; (D)→S					(A) → Q,R ; (B) → P,S ; (C) → S ; (D) → Q,R				
Que.	19									
Ans.	(A) → Q ; (B) → R ; (C) → P ; (D) → S									

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	4	1	3	1	1	1	2	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	1	1	1	2	1	1	3	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	1	1	3	3	3	1	1	2
Que.	31	32	33	34	35	36	37	38	39	
Ans.	3	2	1	B	D	B	A	D	3	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8
Ans.	C	B	T	B	B	B	C, D	9

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